A ratio Lyapunov function for MPRK22(1) with reset generators

Abstract

We study the nonlinear map obtained by applying MPRK22(1) to the conservative system $u' = (\pi \mathbf{1}^T - I)u$, where π is a positive mass-one vector. For every fixed positive step, the actual map takes the closed mass simplex into an explicit compact subset of its interior. On the positive simplex, the classical Hilbert distance to π is a strict Lyapunov function with an explicit contraction factor smaller than one. It follows that every positive mass-one original orbit converges geometrically to π, which is globally asymptotically stable on that simplex. The proof reduces the two linear solves to one scalar function shared by the component ratios and bounds its logarithmic derivative from both sides. The method and projective metric are inherited; the estimate concerns this reset-generator class and this fixed method.

Keywords: Lyapunov function; discrete dynamical system; modified Patankar method; projective metric; global convergence

2020 Mathematics Subject Classification. Primary 65L20; Secondary 37B25, 39A30.

1 Introduction

A Lyapunov function for a numerical recurrence gives information about its actual iterates. For a positive conservative map, the relevant phase space is often a mass simplex. Positivity and mass conservation give boundedness, while a global strict energy can identify the limiting state and quantify attraction. An explicit interior bound is useful because the positive simplex itself is not compact.

Modified Patankar–Runge–Kutta methods preserve positivity and linear conservation laws through state-dependent weights. Their application to a linear differential equation is generally a nonlinear map. Izgin, Kopecz and Meister [1] develop a stability theory for the resulting nonhyperbolic fixed points. For the positive conservative linear systems considered there, they prove local attraction on the matching invariant affine set for MPRK22(α), $\alpha \geq 1/2$, at every fixed positive step. The initial-neighborhood restriction in that theorem is important. Global behavior for nonnegative Runge–Kutta tableaux is discussed as a further question in [2, 3].

We consider one explicit class within that setting. Let $N \geq 2$, let $\pi \in \mathbb{R}^N$ satisfy

$$\pi_i > 0 \quad (1 \leq i \leq N), \qquad \mathbf{1}^T \pi = 1, \qquad A = \pi \mathbf{1}^T - I, \tag{1}$$

and define

$$S = \{u \in \mathbb{R}^N : u_i > 0,\ \mathbf{1}^T u = 1\}, \qquad \overline{S} = \{u \in \mathbb{R}^N : u_i \geq 0,\ \mathbf{1}^T u = 1\}. \tag{2}$$

We use the column-vector convention $u' = Au$. The matrix A is conservative, irreducible and reversible: $\mathbf{1}^T A = 0$ and $a_{ij}\pi_j = a_{ji}\pi_i$. The rank-one matrix is $\pi\mathbf{1}^T$; A has rank $N-1$. We call (1) a reset generator.

For a fixed $h > 0$, the map studied here is

$$\begin{aligned} v &= (I - hA)^{-1}u, \\ w_i &= \frac{u_i + v_i}{2v_i} \quad (1 \le i \le N), \\ T_h(u) &= (I - hA\,\mathrm{diag}(w))^{-1}u. \end{aligned} \tag{3}$$

It is the unrelaxed MPRK22(1) map with the canonical production/destruction split. More explicitly, for $i \neq j$ this split is $p_{ij}(u) = a_{ij}u_j$, $d_{ij}(u) = a_{ji}u_i$, and the diagonal terms are zero. At $\alpha = 1$, equations (25)–(34) of the author version of [1] have

$$B = (I - hA)^{-1}, \qquad C = \tfrac{1}{2}(I + B), \qquad \sigma_i(u) = (Bu)_i, \qquad \tau_i(u) = \frac{(Cu)_i}{\sigma_i(u)} = w_i.$$

Their equation $M(u)u^+ = u$, with $M(u) = I - hA\,\mathrm{diag}(\tau(u))$, is exactly (3). Each stage v_n is computed from the current state u_n; the original orbit has only the initial datum u_0 and satisfies $u_{n+1} = T_h(u_n)$.

The energy in this note is classical Hilbert projective distance; see Nussbaum [4]. Our argument uses a scalar function that is common to all component ratios within one update. It yields an explicit global estimate for (3), including interior capture before convergence is invoked. It does not address the general-generator or general-tableau questions in [2, 3].

Put

$$\begin{aligned} R &= \frac{1}{\min_i \pi_i}, \qquad a = h^2 + 2h + 2, \qquad b = h(h+2), \\ \ell &= \frac{h^2}{aR + b}, \qquad q = \max\left\{\frac{2R}{2R+h}, \frac{aR}{aR+b}\right\}. \end{aligned} \tag{4}$$

The constants are deliberately conservative.

Theorem 1. *Let* (1) *hold and fix* $h > 0$. *Then* $0 < \ell < 1$ *and* $0 < q < 1$, *and the following statements hold.*

(i) *Both systems in* (3) *are nonsingular for every* $u \in S$. *The stage* v *and output* $T_h(u)$ *belong to* S. *Moreover,* T_h *has a continuous extension to* $\overline{S}$ *satisfying*

$$T_h(u)_i \ge \pi_i \ell \quad (u \in \overline{S}, 1 \le i \le N). \tag{5}$$

Thus every allowed original orbit lies from its first step onward in the compact interior set

$$K = \{u : \mathbf{1}^T u = 1,\ u_i \ge \pi_i \ell \text{ for every } i\} \subset S. \tag{6}$$

(ii) *The fixed state function*

$$E(u) = \log \max_i \frac{u_i}{\pi_i} - \log \min_i \frac{u_i}{\pi_i}, \qquad u \in S, \tag{7}$$

is continuous, nonnegative and zero precisely at π. Its finite sublevels are compact in S, and it tends to infinity as u approaches the boundary of $\overline{S}$. For every $u \in S$,

$$E(T_h(u)) \le qE(u). \tag{8}$$

In particular, the decrease is strict whenever $u \neq \pi$.

(iii) *For every $u_0 \in S$, all original iterates are defined and*

$$\begin{aligned} E(u_n) &\le q^n E(u_0), \\ \|u_n - \pi\|_1 &\le \exp\big(q^n E(u_0)\big) - 1 \le q^n\big(\exp(E(u_0)) - 1\big), \qquad n \ge 0. \end{aligned} \tag{9}$$

The equilibrium π is globally asymptotically stable relative to S, and

$$\sum_{n=0}^{\infty} E(u_n) \le \frac{E(u_0)}{1-q}, \qquad \sum_{n=0}^{\infty} \|u_n - \pi\|_1 \le \frac{\exp(E(u_0)) - 1}{1-q}. \tag{10}$$

The factor q depends on π and the fixed step h, but not on the initial state. The estimate is a contraction of the energy relative to π. No contraction estimate between two independently initialized trajectories is asserted. The boundary statement in (5) concerns the continuous extension of the linear-solve formula; the original starts in the theorem remain strictly positive.

2 Proof of Theorem 1

We first solve the two systems and establish interior capture. Then a scalar logarithmic derivative estimate gives the strict energy inequality.

Lemma 1. *For every $u \in \overline{S}$, the formulas* (3) *define a unique positive mass-one output. If $x_i = u_i/\pi_i$, there is a common scalar $z \ge h/2$ such that*

$$\frac{T_h(u)_i}{\pi_i} = F_z(x_i), \qquad F_z(x) = \frac{2(x+h)(x+z)}{ax+b}. \tag{11}$$

The map is continuous on $\overline{S}$ and satisfies (5).

Proof. Let $P = \pi \mathbf{1}^T$. Since $P^2 = P$, direct multiplication gives

$$(I - hA)^{-1} = \frac{I + hP}{1+h}, \qquad v = \frac{u + h\pi}{1+h}.$$

This proves the first nonsingularity statement, positivity of v and $\mathbf{1}^T v = 1$. Also

$$\begin{aligned} 0 \le x_i \le R, \qquad \sum_i \pi_i x_i = 1, \qquad w_i &= \frac{(h+2)x_i + h}{2(x_i+h)}, \\ w_i - \frac{1}{2} = \frac{(h+1)x_i}{2(x_i+h)} \ge 0, \qquad d_i := 1 + hw_i &= \frac{ax_i + b}{2(x_i+h)} > 0. \end{aligned} \tag{12}$$

All denominators are positive, including at a boundary input.

Write $D = \operatorname{diag}(d_i)$. The second coefficient matrix is $D - h\pi w^T$. Its rank-one denominator is

$$\Delta = 1 - hw^T D^{-1}\pi = \sum_i \frac{\pi_i}{d_i} > 0, \tag{13}$$

where $hw_i/d_i = 1 - 1/d_i$ was used. Hence

$$(D - h\pi w^T)^{-1} = D^{-1} + \frac{h}{\Delta} D^{-1}\pi w^T D^{-1}. \tag{14}$$

For example, multiplication by $D - h\pi w^T$ verifies this identity because the remaining rank-one coefficient is $-h + h/\Delta - h^2(w^T D^{-1}\pi)/\Delta = 0$.

Every entry of the second term in (14) is strictly positive. Since $u \geq 0$ has mass one, its image $t = T_h(u)$ is strictly positive. Multiplication of the defining equation by $\mathbf{1}^T$ gives

$$\mathbf{1}^T t = \mathbf{1}^T (I - hA\operatorname{diag}(w))t = \mathbf{1}^T u = 1.$$

This proves positivity and mass conservation before any lower output bound is assumed.

Now define $z = hw^T t$. By (12), $z \geq h/2$. The second system is $Dt = u + \pi z$, which gives exactly (11). An expression involving only the input is

$$z = h\frac{\sum_i w_i u_i/d_i}{\sum_i \pi_i/d_i}. \tag{15}$$

Its numerator is positive since at least one u_i is positive. Equations (12)–(15) prove continuity on $\overline{S}$ and agree with the original map on S, giving the claimed continuous extension.

Finally, $x_i \geq 0$, $x_i \leq R$ and $z \geq h/2$ imply

$$\frac{t_i}{\pi_i} = \frac{2(x_i + h)(x_i + z)}{ax_i + b} \geq \frac{h^2}{aR + b} = \ell.$$

Since $R \geq N \geq 2$ and $a > h^2$, we have $0 < \ell < 1$. Thus K is nonempty, closed, bounded and contained in S; it contains π. The inclusion $T_h(\overline{S}) \subseteq K$ also makes K positively invariant. This establishes capture before any convergence argument. ∎

Lemma 2. *Let $z \geq h/2$ and define F_z by (11). For all $x, y \in (0, R]$,*

$$|\log F_z(x) - \log F_z(y)| \leq q|\log x - \log y|. \tag{16}$$

Proof. Logarithmic differentiation gives

$$\frac{xF_z'(x)}{F_z(x)} = \frac{x}{x + h} + \frac{x}{x + z} - \frac{ax}{ax + b}. \tag{17}$$

The relation $ah - b = h^2(h + 1) > 0$ implies

$$\frac{x}{x + h} - \frac{ax}{ax + b} = -\frac{h^2(h + 1)x}{(x + h)(ax + b)} < 0.$$

Consequently, for $0 < x \le R$,

$$\frac{xF_z'(x)}{F_z(x)} \le \frac{x}{x+z} \le \frac{x}{x+h/2} \le \frac{2R}{2R+h}. \tag{18}$$

The first two terms of (17) are positive, so

$$\frac{xF_z'(x)}{F_z(x)} \ge -\frac{ax}{ax+b} \ge -\frac{aR}{aR+b}. \tag{19}$$

Both positive fractions in (4) are strictly less than one, giving $0 < q < 1$. Equations (18) and (19) bound the absolute derivative of $s \mapsto \log F_z(e^s)$ by q on every interval with $e^s \in (0, R]$. The mean value theorem proves (16). The argument allows F_z' to have either sign. ∎

Proof of Theorem 1. Part (i) and the bounds on ℓ, q follow from the two lemmas. For $u \in S$, put

$$m = \min_i \frac{u_i}{\pi_i}, \qquad M = \max_i \frac{u_i}{\pi_i}.$$

The weighted-mean relation $\sum_i \pi_i(u_i/\pi_i) = 1$ gives $0 < m \le 1 \le M$. Thus $E(u) = \log(M/m) \ge 0$, and equality holds exactly when every ratio is one. Continuity follows from positivity and the continuity of a finite maximum and minimum.

If u approaches the boundary of $\overline{S}$, then m tends to zero while $M \ge 1$, so $E(u) \ge -\log m$ tends to infinity. More explicitly, $E(u) \le c$ for a finite $c \ge 0$ implies $m \ge e^{-c}$. The corresponding sublevel is closed in the compact set

$$\{u \in \overline{S} : u_i \ge \pi_i e^{-c} \text{ for every } i\} \subset S,$$

and is therefore compact. Sublevels with $c < 0$ are empty.

For a single input u, Lemma 1 gives the same z in every component of its output. Apply Lemma 2 to each pair $x_i = u_i/\pi_i$, $x_j = u_j/\pi_j$. It follows that

$$\left|\log \frac{T_h(u)_i}{\pi_i} - \log \frac{T_h(u)_j}{\pi_j}\right| \le q|\log x_i - \log x_j| \le qE(u).$$

Choosing indices that realize the largest and smallest output ratios proves (8), without a coordinate-ordering assumption. Since $q < 1$, the decrease is strict away from π.

All iterates exist by Lemma 1, and induction in (8) yields the first estimate in (9). The relations $m \le 1 \le M$ and $M/m = e^{E(u)}$ also give

$$e^{-E(u)} \le \frac{u_i}{\pi_i} \le e^{E(u)}, \qquad \|u - \pi\|_1 \le e^{E(u)} - 1. \tag{20}$$

For the norm estimate, use $1 - e^{-E} \le e^E - 1$ and sum the component bounds with weights π_i. Convexity of the exponential gives

$$e^{tE_0} - 1 \le t(e^{E_0} - 1) \quad (0 \le t \le 1,\ E_0 \ge 0).$$

Taking $E_0 = E(u_0)$ and $t = q^n$ proves the remaining estimates in (9). Summing these nonnegative geometric bounds proves (10). In particular, every allowed orbit converges to π.

We verify Lyapunov stability as well. Given $\varepsilon > 0$, set

$$\delta = \frac{\varepsilon}{R(2+\varepsilon)}.$$

If $d_0 = \|u_0 - \pi\|_1 < \delta$, then $|u_{0,i}/\pi_i - 1| \le Rd_0 < 1$ and hence

$$e^{E(u_0)} - 1 \le \frac{1 + Rd_0}{1 - Rd_0} - 1 = \frac{2Rd_0}{1 - Rd_0} < \varepsilon.$$

Equation (9) gives $\|u_n - \pi\|_1 < \varepsilon$ for every $n \ge 0$. Thus π is stable relative to S, and the attraction just proved makes it globally asymptotically stable there. At $u_0 = \pi$, the two solves give $v_n = \pi$, $w_n = \mathbf{1}$ and $u_n = \pi$ for every n; all zero-energy cases are included. ∎

3 Scope of the estimate

The function (7) is classical projective distance. General derivative criteria for projective Lipschitz maps, such as [4, Theorem 2.4], concern the full derivative of a map and comparisons between two inputs. Here the scalar z is held fixed only while comparing components belonging to one actual update. It generally changes when the entire input changes. The proof consequently establishes the stated one-point Lyapunov inequality, without importing a stronger two-input premise.

The generator class and canonical split are essential to the scalar representation (11). We retain the initial-state restriction and method parameter exactly. The theorem is separate from the gamma-dependent positivity-relaxation procedures in [5], and it makes no assertion about a general prescribed thermodynamic free energy. For example, the nonlinear reaction-wise splitting in [6, Section 6.2] gives a different map and a different functional, including reported free-energy increases for MPRK22(1). Those examples neither supply nor contradict the present reset-map estimate.

The source comparison uses the author version of [1], the published later discussion [2], the stated thesis revision [3], and the specified recent preprints. It is a bounded comparison, and no claim of firstness or a general global-conjecture resolution is made. A companion verification note records the exact source locators and version limits. The accompanying exact-check program verifies the rational identities and representative literal solves; the all-state and all-index conclusions follow from the proof above.

References

[1] T. Izgin, S. Kopecz and A. Meister, On the stability of unconditionally positive and linear invariants preserving time integration schemes, *SIAM J. Numer. Anal.* **60** (2022), 3029–3051. Author version: arXiv:2202.11649v1.

[2] T. Izgin, S. Kopecz, A. Meister and A. Schilling, On the non-global linear stability and spurious fixed points of MPRK schemes with negative RK parameters, *Numer. Algorithms* **96** (2024), 1221–1242.

[3] T. Izgin, *A unifying theory for Runge–Kutta-like time integrators: convergence and stability*, Ph.D. thesis, University of Kassel, 2024. Revised author version: arXiv:2402.13788v3, 2025.

[4] R. D. Nussbaum, Finsler structures for the part metric and Hilbert's projective metric and applications to ordinary differential equations, *Differential Integral Equations* **7** (1994), 1649–1707.

[5] T. Izgin, H. Ranocha and C.-W. Shu, A positivity-preserving relaxation algorithm, arXiv:2604.02308v1, 2026.

[6] J. Zhang, Layer-wise discrete certificates in multicomponent reacting-flow discretizations, arXiv:2609.27272v1, 2026.